Fair Allocation in Flow Games

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Model Definitions

Flow Games (Single Edge per Player)
The system is a directed graph $G = (V, E)$ with source $s \in V$ and sink $t \in V$.
Each player $i \in \{1, \ldots, N\}$ controls a single edge $e_i \in E$. Each edge e_i has capacity $c_i \geq 0$.
Characteristic Function For any coalition $S \subseteq N$,

$$v(S) = \text{maximum } s\text{-}t \text{ flow using only edges controlled by } S.$$

Core

An allocation vector $x \in \mathbb{R}^n$ is in the core if it is efficient ($\sum_{i \in N} x_i = v(N)$) and coalitionally rational:

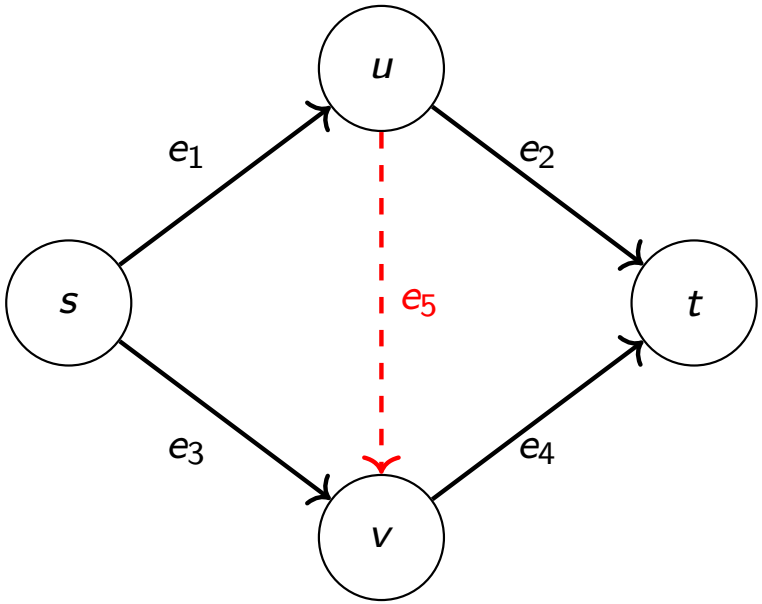
$$\sum_{i \in S} x_i \geq v(S) \quad \forall S \subseteq N$$

Shapley Value
The Shapley value ϕ allocates payoffs based on the expected marginal contribution of player i over all random arrival permutations:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S))$$

Shapley not in Core

If an edge does not belong to any minimum s-t cut, then it gets payoff 0 in every core allocation. But, for Shapley value calculation the edge gets a positive Shapley value if it belongs to any s-t cut.



Instance:

- All capacities $= 1$
- Max flow $v(N) = 2$
- Minimum cuts: $\{e_1, e_3\}$, $\{e_2, e_4\}$

Key point:

- e_5 is not in any minimum cut
- $\Rightarrow$ Core payoff: $x_5 = 0$
- But e_5 lies in some s-t cuts
- $\Rightarrow$ Shapley value $\phi_5 > 0$

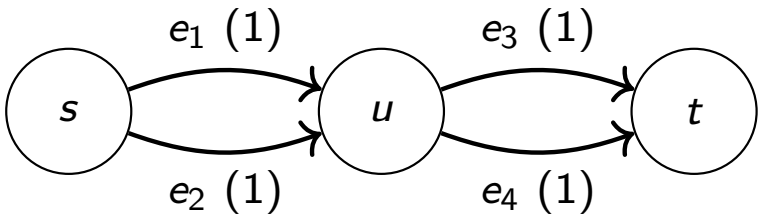
Shapley in Core

Convex games
Condition:

$$v(S \cup \{i\}) - v(S) \leq v(T \cup \{i\}) - v(T) \tag{1}$$

for all $S \subseteq T \subseteq N \setminus \{i\}$. In a flow game, this implies that an edge i becomes more valuable as the set of available edges grows, as it may complete more $s - t$ paths.
Shapley('71) showed that Shapley value lies at the centre of the core for Convex games.

Non convex game



Max flow: $v(N) = 2$

Shapley values:

- $\phi(e_1) = 1/2$, $\phi(e_2) = 1/2$
- $\phi(e_3) = 1/2$, $\phi(e_4) = 1/2$

The above example the Shapley value is in the core.

Introducting SP balanced networks

For now we limit our discussions to graphs with all edge weights equal to 1. An SP-balanced network is defined as a series composition of parallel layers, where each parallel circuit contains an identical number of edges. Let x denote the number of layers connected in series, and y the number of parallel edges in each layer. By symmetry, each edge has identical marginal contributions across permutations, and the Shapley value resolves uniformly as $\phi_e = 1/x$.

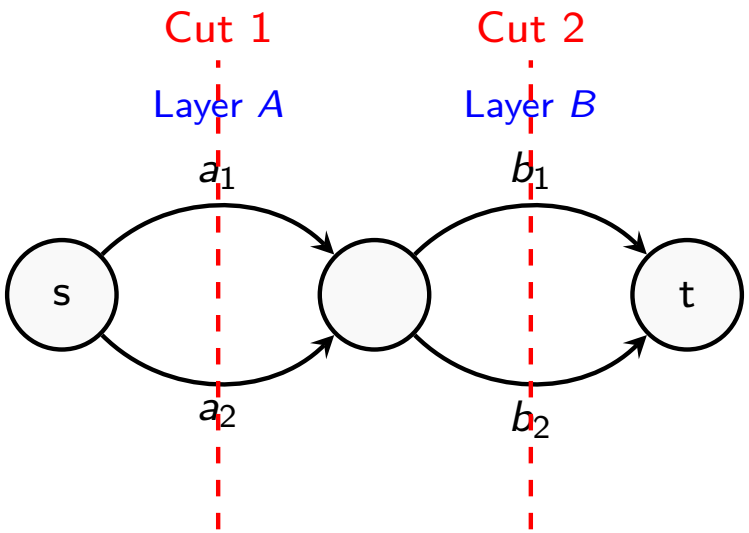


Figure: A 2×2 SP-balanced network. Note that any $s - t$ path requires one edge from Layer A and one from Layer B.

Crucially, any minimal winning coalition must include at least one edge from each of the x layers. Therefore, the total payoff assigned to such a coalition is exactly

$$\sum_{e \in P} \phi_e = x \cdot \frac{1}{x} = 1,$$

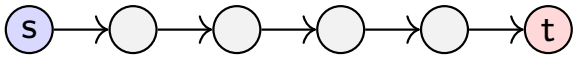
Experimental results

All simulations were run using a custom exact-arithmetic Python framework. Maximum flows were calculated via the Edmonds-Karp algorithm. To avoid floating-point inaccuracies during core deficit calculations, all Shapley marginal contributions were evaluated using Python's `fractions` library.

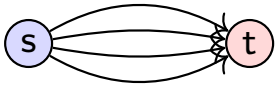
Table: Selected Topological Results ($c_i = 1$)

Topology	n	**Convex?**	**In Core?**	**Deficit**
Path-5	5	Yes	Yes	0.000
Parallel-4	4	Yes	Yes	0.000
SP-Balanced (2×2)	4	Yes	Yes	0.000
Series(3)ofParallel(2)	6	**No**	**Yes**	0.000
Series(2)ofParallel(3)	6	**No**	**Yes**	0.000
Bottleneck (Funnel)	5	No	No	0.266
Double Bottleneck	9	No	No	0.330
Wheatstone Bridge	5	No	No	0.033
FanIn ($3 \to 1$)	7	No	No	0.304

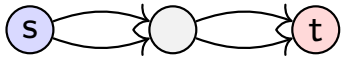
Graphs



(1) Path-5

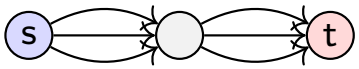


(2) Parallel-4

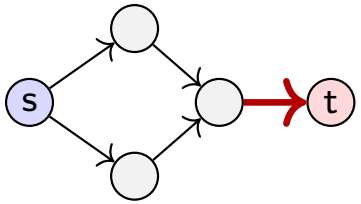


(3) SP-Bal. 2×2

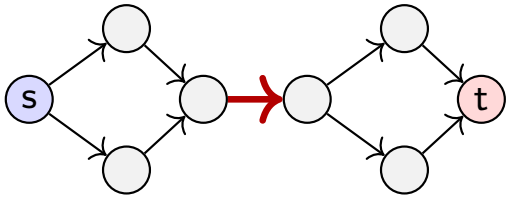
(4) Series(3)ofPar(2)



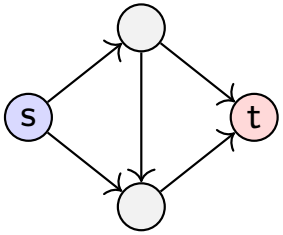
(5) Series(2)ofPar(3)



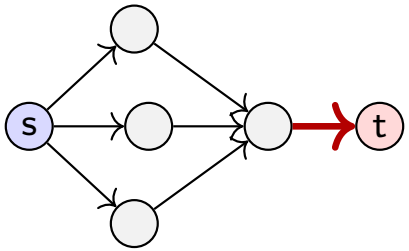
(6) Funnel (Bottleneck)



(7) Double Bottleneck



(8) Wheatstone Bridge



(9) FanIn(3→1)

Figure: The nine network topologies evaluated. Blue = source s, red = sink t, thick red edges = bottleneck (single min-cut edge). Topologies (1)–(5) have Shapley in core; (6)–(9) do not.

Analysis of results

All simulations were run using a custom exact-arithmetic Python framework. Maximum flows were calculated via the Edmonds-Karp algorithm.
Results support our topological hypotheses:

1. **Paths & Parallels:** Simple series and parallel networks are convex. Marginal contributions are monotone, so the Shapley value lies in the core. Every minimal s-t path receives total payoff ≥ 1.

2. **Series of Parallels:** Graphs like `Series(3)ofParallel(2)` are *not convex* but still stable. They satisfy SP-Balanced structure $\Rightarrow$ zero core deficit. Symmetry across layers equalizes contributions, so all path constraints hold.

3. **Bottleneck Failures:** Asymmetric graphs (e.g., `Bottleneck`, `FanIn(3->1)`) violate the core. Critical edges receive excessive payoff (e.g., $\phi = 7/15$), while paths get < 1 (e.g., $11/15$).

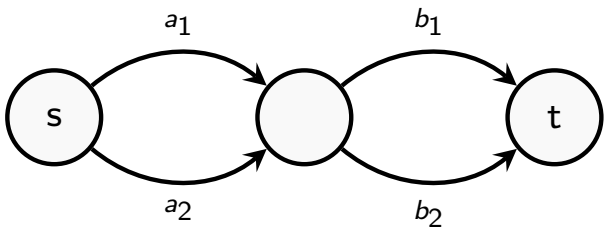
   $\Rightarrow$ *Bottleneck tax:* instability due to uneven allocation.

What if Shapley doesn't lie in the Core?

The Core is a Polytope, not a Point
When $\phi \notin \mathcal{C}(v)$, we look for the "most stable" allocation *inside* the core.
The core is a convex polytope — infinitely many valid allocations exist.

Example: 2×2 SP-balanced network (a_1, a_2, b_1, b_2), $v(N) = 1$.
Both of the following are valid core allocations:

$$x^{(1)} = (1,\ 0,\ 1,\ 0), \qquad x^{(2)} = \left(\tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2}\right)$$



$x^{(1)} = (1, 0, 1, 0)$: stable, but assigns **zero** to a_2, b_2 despite being structurally identical to a_1, b_1.

$x^{(2)} = (0.5, 0.5, 0.5, 0.5)$: treats all players equally *and* sits **deeper** inside the core.

Key Question

Among all valid core allocations, which one is the *most stable* and *most equitable*?

The Nucleolus: Deepest Point in the Core

The Nucleolus [?]

Define the *excess* of coalition S under allocation x:

$$e(S, x) = v(S) - \sum_{i \in S} x_i$$

The nucleolus η lexicographically minimises the sorted (descending) excess vector $\theta(x)$ over all $x \in \mathcal{C}(v)$:

$$\eta = \arg \min_{x \in \mathcal{C}(v)} \theta(x)$$

Intuitively: η makes the *most dissatisfied* coalition as satisfied as possible, resolves ties for the next, and so on. It is the point sitting **deepest inside the core**.

SP-balanced case: For the 2×2 network, the nucleolus evaluates to:

$$\eta = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right) = \phi$$

The Shapley value *coincides* with the nucleolus — it is both fair *and* maximally stable.

Asymmetric networks (e.g. Funnel):
The nucleolus assigns $\eta = (0, 0, 1)$, concentrating all payoff on the bottleneck.
The Shapley value gives $\phi = (1/6, 1/6, 2/3)$, spreading value to redundant edges.
In further experiments, alongside the core deficit, we also measure the **Shapley–Nucleolus L_1 distance** $\|\phi - \eta\|_1 = \sum_i |\phi_i - \eta_i|$

Bottleneck Capacity: Phase Transition to Core Stability

Setup: Funnel with $k = 2$ (13) parallel entry edges (unit capacity) and a single bottleneck edge of capacity c_b. How does core membership change as c_b varies?

c_b	Deficit	$\|\phi - \eta\|_1$	In Core?
0.50	0.133	0.533	No
0.75	0.200	0.800	No
1.00	0.267	1.067	No
1.50	0.000	0.533	**Yes**
2.00	0.000	0.000	**Yes**

Two key observations:

- **Deficit is non-monotone:** it *rises* as c_b goes from $0.5 \to 1.0$ before collapsing to zero at $c_b = 1.5$. Expanding the bottleneck increases grand-coalition flow, but worsens mis-attribution to redundant entry edges.
- **Sharp phase transition** at $c_b = k$: strictly outside the core for $c_b < k$, strictly inside for $c_b \geq k$.

Interpretation

At $c_b = k = 2$, both layers have equal capacity $\Rightarrow$ network becomes SP-balanced $\Rightarrow \phi = \eta$ and deficit vanishes simultaneously. The minimum investment to stabilise the network is to expand the bottleneck to match the total entry capacity.

Edge Capacity Asymmetry: Deficit and Nucleolus Distance Decouple

Setup: Two-layer bipartite network (13). Entry and cross edges have unit capacity. Exit edges are imbalanced by α: $b_1 \to t$ has capacity $1+\alpha$, $b_2 \to t$ has capacity $1-\alpha$. Total exit capacity fixed at 2.

α	Deficit	$\|\phi - \eta\|_1$	In Core?
0.0	0.224	2.448	No
0.2	0.090	1.779	No
0.4	0.067	1.334	No
0.6	0.045	0.979	No
0.7	0.034	1.100	No
0.8	0.022	2.178	No
0.9	0.011	2.022	No

Two key observations:

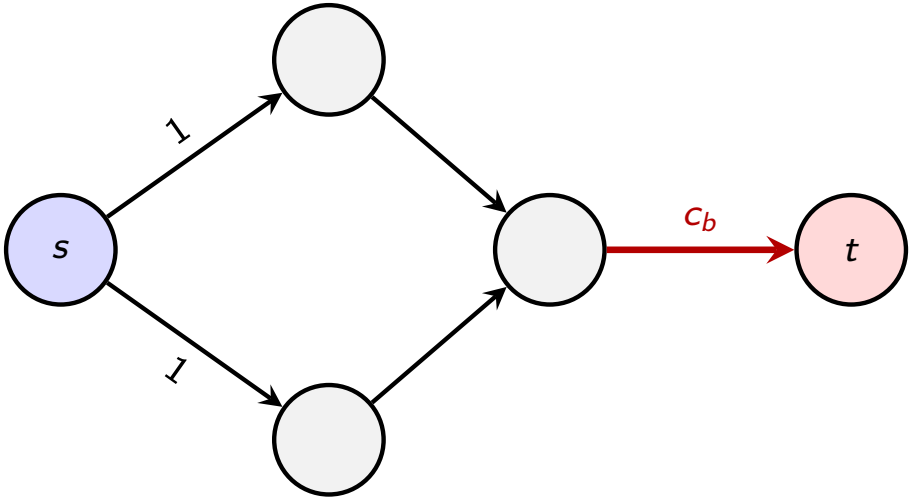
- **Deficit decreases monotonically.** As α grows, $b_1 \to t$ dominates and the network increasingly resembles a funnel. The Shapley value concentrates more payoff on the dominant edge, shrinking the path-condition violation.
- $\|\phi - \eta\|_1$ **is non-monotone.** It reaches a minimum near $\alpha = 0.6$ before rising again. The nucleolus tracks the capacity ratio of exit edges exactly, while the Shapley value tracks the same direction but more slowly, averaging over all permutations.

Key Insight: Stability and Fairness are Orthogonal

As α increases, the network moves toward stability (deficit $\downarrow$) but simultaneously drifts away from the most equitable allocation ($\|\phi - \eta\|_1 \uparrow$). Achieving both requires SP-balanced structure.

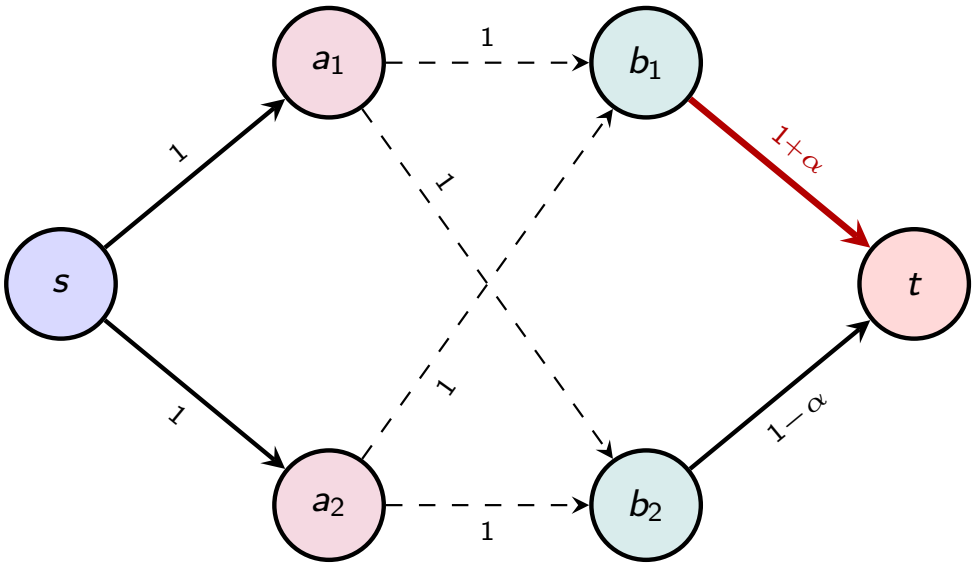
Network Topologies for Capacity Experiments

Funnel ($k = 2$, varying c_b)



Bottleneck edge capacity c_b varies.
Entry edges fixed at unit capacity.
Phase transition at $c_b = k = 2$.

Bipartite Mixed-Layer (varying α)



Exit layer imbalanced by $\alpha \in [0, 1)$.
Total exit capacity fixed at 2.
Deficit $\downarrow$ monotonically; $\|\phi - \eta\|_1$ non-monotone.

Converting to SP-Balanced Form: Greedy Bypass Surgery

Motivation: If SP-balanced networks guarantee stability, can we restructure an unstable network into SP-balanced form through minimal interventions?

Intervention: Add a single public bypass edge alongside the highest-frequency edge (the edge appearing in the most minimum cuts). The player set remains fixed; only the characteristic function changes:

$$v'(S) = \text{max-flow in } G' \text{ using edges } S \cup E_{\text{bypass}}$$

Topology	σ_0^2	σ_f^2	**Deficit**$_0$	**Deficit**$_f$
Funnel(2+1)	0.160	0.000	0.267	0.100
Funnel(3+1)	0.122	0.000	0.305	**0.562** ↑
Double-Bottleneck	0.099	0.000	0.330	0.056
Wheatstone-Bridge	0.204	0.000	0.348	0.195
Ladder-2×2	2.160	0.000	0.031	**1.031** ↑

σ_0^2, σ_f^2 = cut-size variance before/after surgery. Deficit$_0$, Deficit$_f$ = core deficit before/after.

Key Finding: $\sigma^2 = 0$ does not guarantee deficit reduction

For Funnel(3+1) and Ladder-2×2, bypass surgery *worsens* the deficit. Bypassing the bottleneck expands grand-coalition flow, but if the entry layer is narrower than the expanded bottleneck, the entry edges become the new binding constraint — instability shifts upstream rather than being removed.

Conclusion: True stabilisation requires matching bypass width to *every* layer simultaneously — local surgery is insufficient. Converting an arbitrary network to SP-balanced form requires global redesign, not incremental edge additions.

Other Directions Explored

Cut-Size Variance as a Predictor of $\|\phi - \eta\|_1$
We computed σ^2, the variance in sizes of minimum cuts across the network, and examined its correlation with the Shapley–Nucleolus L_1 distance. The intuition: higher variance in cut sizes implies more asymmetric marginal contributions, which should push ϕ further from η.

Correlation Metrics:
To quantify this relationship we computed two standard correlation coefficients:

$$r_{\text{Pearson}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}, \qquad r_{\text{Spearman}} = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

where d_i is the difference in ranks of each observation.

- $r_{\text{Pearson}} = 0.04$ — near zero, indicating the relationship between σ^2 and $\|\phi - \eta\|_1$ is **not linear**
- $r_{\text{Spearman}} = 0.88$ — strong rank correlation, suggesting the relationship is **monotone but step-like**: $\|\phi - \eta\|_1$ jumps in discrete stages as σ^2 crosses certain thresholds rather than increasing smoothly

Dataset Bias

Out of 23 topologies tested, 14 had $\|\phi - \eta\|_1 = 0$, heavily skewing the dataset toward the trivial case. The observed correlations may not generalise — further experimentation on a more balanced set of topologies is needed before drawing firm conclusions.

PJM Interconnection and Transmission Cost Allocation

Challenge: multiple utilities owned different transmission lines, while electricity flowed across a shared, highly interconnected grid. The problem was how to allocate congestion revenues and transmission costs in a way that was both fair and stable.
Modeling the grid as a cooperative flow game, the Shapley value provides a fairness-based allocation rule. However, PJM's meshed topology (with loops and alternative paths) creates instability: some lines are critical in small coalitions but become replaceable in larger ones, reducing their marginal contribution. This violates convexity and can push the Shapley value outside the core:

$$\sum_{e \in P} \phi_e < 1.$$

In practice, this manifests as disputes over tariffs and cost recovery rather than physical exit.
PJM's solutions, though not explicitly game-theoretic, mirror series-parallel (SP) simplifications:

- **Contract path approximation:** treats flows as if they follow simple paths, clarifying coalition structure.
- **Zonal aggregation:** groups lines into regions, reducing asymmetries and treating substitutes as parallel components.
- **Reduced emphasis on redundant cross-ties:** limits the destabilizing impact of alternative paths.

These interventions effectively simplify the network toward a series-parallel structure, stabilizing marginal contributions and making fair allocations more sustainable.
Takeaway: Practical market design naturally moves toward SP-like structures because they improve stability, suggesting that transmission cost allocation could be systematically improved using SP-balancing principles.

Horizontal Alliances in Logistics

- **Setting:** In Less-Than-Truckload (LTL) horizontal alliances, competing carriers pool freight to improve utilization and reduce costs.
- **Model:** The system can be modeled as a cooperative flow game:
  - players: carriers,
  - value: achievable cost savings,
  - allocation: Shapley value as a fairness-based rule.
- **Bottleneck Issue:** Centralized consolidation hubs create funnel-like structures:
  - hubs act as near-veto players,
  - required for most feasible routes,
  - receive disproportionately high marginal contributions.
- **Core Instability:**
  - allocations become highly skewed toward the hub,
  - alternative routing options enable sub-coalitions to bypass the hub,
  - some coalitions receive less than their standalone value,
  - leads to violations of core constraints and incentives to defect.
- **Stabilization Strategy:** Modify network structure rather than only allocation rules:
  - introduce multiple parallel hubs with comparable capacity,
  - distribute flow across hubs,
  - reduce dependence on any single bottleneck.
- **SP-Balanced Interpretation:**
  - transforms the network toward a series–parallel structure,
  - parallel hubs add redundancy and limit dominance,
  - reduces variability in marginal contributions,
  - mitigates cut-switching behavior.
- **Outcome:**
  - improves stability of allocations,
  - increases likelihood that the Shapley value lies in the core,
  - enhances robustness of cooperative agreements.

Conclusion

- **Key Insight:** Stability of the Shapley value in flow games depends primarily on *network topology*, not just convexity.
- **SP-Balanced Networks:**
  - perfectly balanced series–parallel (SP) networks ensure both fairness and stability,
  - this holds even in non-convex settings.
- **Asymmetric Structures:**
  - bottlenecks and redundant layers create the *bottleneck tax*,
  - leads to violations of core constraints.
- **Computational Findings:**
  - path-based conditions effectively characterize core membership,
  - provide a simple and practical stability check.
- **Shapley Value vs Nucleolus:**
  - in SP-balanced networks, Shapley value = symmetric nucleolus,
  - in asymmetric networks the two diverge significantly, highlighting trade-offs.
- **Role of Nucleolus:**
  - minimizes dissatisfaction of worst-off coalitions,
  - provides a stronger notion of fairness.
- **Bottleneck Tax Interpretation:**
  - divergence between Shapley value and nucleolus measures inefficiency,
  - captures trade-off between probabilistic fairness and stability.
- **Practical Implications:**
  - relevant to electricity markets and logistics alliances,
  - SP-balancing improves robustness and reduces instability.
- **Future Work:**
  - extend analysis to large meshed networks,
  - design scalable mechanisms combining topology, fairness, and stability.

Code & Repository

All code and experiments are publicly available at:

https://github.com/Dheeraj-hash/Stability_vs_Fairness_in_flow_games